

Student Performance Analysis Report

Artificial Intelligence, Machine Learning & Deep Learning

Student Name: Muhammad Umar

Dataset: students_performance_dataset_500.csv

Student ID: 09

Role: Junior AI Engineer (Trainee)

1. Scenario & Objective

As a Junior AI Engineer, the primary objective of this assignment is to explore, clean, and analyze a comprehensive student performance dataset comprising 500 students across 11 distinct features before training predictive machine learning models.

2. Implementation Methodology & Code Overview

The analysis was conducted using Python with **Pandas** for data manipulation and inspection, and **NumPy** for numerical array computations. Below is the structured implementation summary:

- Data Loading & Inspection:** Loaded the dataset into a Pandas DataFrame, verified data dimensions (500 rows, 11 columns), and inspected head/tail records.
- Exploratory Data Analysis:** Extracted core categorical insights, calculating total student count, unique departments, and gender demographics.
- Selection & Filtering:** Applied column selection (`df["Column"]`) and conditional filtering (e.g., `Final_Marks > 40`, `Attendance > 90%`).
- NumPy Array Operations:** Converted `Final_Marks` into a NumPy array to compute statistical metrics (`μ`, `min`, `max`) and perform element-wise transformations (`sqrt(x)`, `2x`).

3. Key Findings & Analysis Summary

Through systematic data exploration and vectorised NumPy calculations, the following foundational insights were uncovered:

- Total Cohort Size:** Exactly 500 student records analyzed across 5 academic departments.
- Attendance Trends:** Overall student attendance averages 77.07%, highlighting areas where targeted academic intervention may be beneficial.
- Examination Performance:** Final marks range from a minimum of 20 to a maximum of 50, with an overall average final score of 35.08.

4. Required Program Output

The final script execution successfully compiled and verified the required student performance summary report as shown in the console output below:

```
... =====  
STUDENT PERFORMANCE REPORT  
=====  
Total Students: 500  
Total Departments: 5  
Average Attendance: 77.07%  
Average Final Marks: 35.08  
Highest Final Marks: 50  
Lowest Final Marks: 20  
=====
```